

Real-Time Object Tracking using Mean Shift and CAMShift

1. Introduction

Object tracking is an important thing in computer vision. This means tracking the motion of an object in consecutive images of a video. Applications range from surveillance systems, driverless cars, to hand gesture recognition systems.

Mean Shift is a nonparametric technique that aims at locating the mode of a probability distribution function. In object tracking applications, it creates a color histogram for the target and moves a window to the region that best matches its distribution. CamShift (Continuously Adaptive Mean Shift) is an extension of Mean Shift. The difference is that CamShift can also resize and rotate the tracking window, which helps when the object changes size or angle.

2. Methodology

2.1 Mean Shift Algorithm

The Mean Shift algorithm was first introduced in the paper by Comaniciu and Meer (2002). The feature space can be thought of as a probability density map. Areas with more data points are denser, and those dense areas are called modes. Mean Shift works by placing a window somewhere and moving it step by step toward whichever direction gets denser. The algorithm is carried out until it reaches a certain stage at which it comes to an end, with the final state being called the mode. It is guaranteed to always reach convergence, that is, it will never get stuck in a loop. Only one parameter has to be specified for this algorithm h .

The steps for using Mean Shift in tracking are:

- Select the target object in the first frame and compute its HSV colour histogram.
- For each new frame, create a back-projection image where each pixel value shows how likely it belongs to the target.
- Run Mean Shift on the back-projection starting from the last known window position.
- The window shifts toward the highest colour density and stops when it converges.
- That final position is the new location of the object.

2.2 CamShift Algorithm

CamShift wraps Mean Shift with an extra step that adjusts the window each frame:

- Run Mean Shift to get the converged location.
- Calculate the zeroth and second spatial moments of the result.

- Resize the window based on the zeroth moment (total probability mass).
- Estimate the orientation and return a rotated bounding ellipse.

This makes CamShift better than basic Mean Shift when the object changes size or rotates. Mean Shift keeps a fixed window, but CamShift updates it every frame.

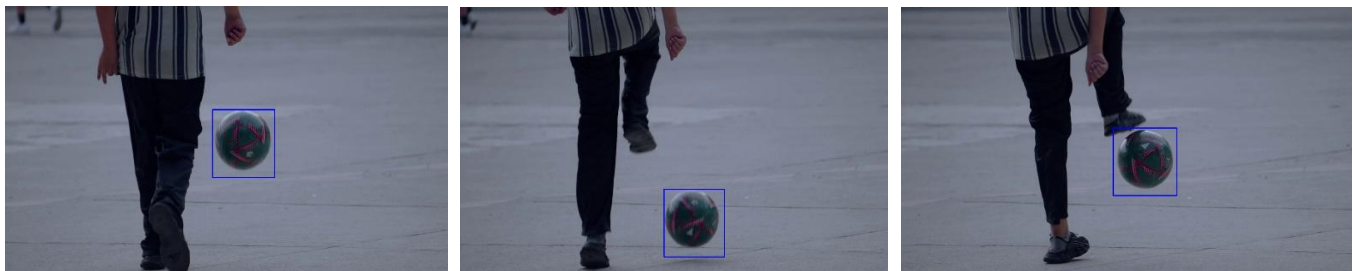
2.3 Implementation

Both algorithms were tested using Python and OpenCV (cv2). Here is what was done:

- The target (football) was selected manually in the first frame.
- HSV hue histogram (180 bins) calculation for the selected region of interest (ROI).
- Application of `cv2.calcBackProject()` function to generate back projection image.
- Use of either `cv2.meanShift()` or `cv2.CamShift()` function with stopping criteria of maximum iterations or displacement of 1 pixel.
- The tracking window was drawn on each frame and selected frames were saved.

3. Results

3.1 Mean Shift Tracking Results



Mean Shift Results:

The football was tracked successfully using Mean Shift (shown with a blue bounding box). The window followed the ball as it moved around the frame. But because the window size stays fixed, it could not adjust when the ball appeared larger or smaller at different distances. This is a known limitation of the basic Mean Shift approach.

3.2 CamShift Tracking Results



CamShift Results:

CamShift was tested on the same sequence (green bounding box). It tracked the ball along a similar path but showed a clear advantage, the window adjusted its fit to the object better across frames. This confirms that CamShift handles scale and orientation changes better than basic Mean Shift.

4. Discussion

a) What assumptions does K-Means make?

K-Means assumes:

- The number of clusters (K) must be set before running.
- Clusters are circular or spherical in shape.
- All clusters are roughly the same size.
- Data is evenly spread within each cluster.

b) What is mode-seeking in Mean Shift?

The mode is the position of highest probability in the probability distribution. In mode seeking, the task of the algorithm is to find this position by:

- Starting at an initial window position.
- Computing the mean shift vector, the direction toward the densest nearby region.
- Moving the window in that direction.

This keeps repeating until the window can no longer move to a denser region, that is where the mode is.

c) Does Mean shift require specifying K?

No. Mean Shift finds the number of clusters on its own based on the density of the data. This is one of its biggest advantages when comparing with K-Means.

d) What does bandwidth control?

Bandwidth sets the size of the kernel window, how far the algorithm looks when calculating the local density.

- Small bandwidth - many small & detailed clusters

- Large bandwidth - fewer & broader clusters

It is very important to select the right bandwidth because it directly affects what clusters the algorithm finds.

e) Advantages and disadvantages of Mean Shift vs K-Means

Mean Shift

- No need to set K
- Can handle irregular, complex shapes
- Slow on large datasets
- Robust to outliers
- Automatically finds the number of clusters

K-Means

- Must be manually defined (K)
- Only handles spherical/convex shapes
- Very fast and scalable
- Very sensitive to outliers
- Requires K to be pre-specified

f) Why does modern CV still use K-Means?

Even though Mean Shift gives better results on complex data K-Means is still used because,

- It runs much faster and works on large datasets.
- It is simple to code and easy to scale up.
- In many real-world cases, speed is more important than perfect accuracy.

Non-parametric methods like Mean Shift do not scale well when the feature space has many dimensions. That is why faster and simpler methods like K-Means are still preferred in large-scale computer vision systems.

5. Conclusion

This report looked at Mean Shift and CamShift for object tracking. Mean Shift finds the mode of a colour distribution and moves a window toward it frame by frame. CamShift improves on this by also adjusting the window size and angle.

Testing on a football sequence showed that Mean Shift tracked the ball well but could not handle scale changes. CamShift adapted better and gave a tighter fit to the object.

So when we comparing K-Means and Mean Shift, Mean Shift is more flexible and does not need K to be set. But K-Means is still often used because it is fast and simple.

References

Comaniciu, D. & Meer, P. (2002). "Mean Shift: A Robust Approach Toward Feature Space Analysis." IEEE Transactions on Pattern Analysis and Machine Intelligence, 24(5), p. 603-619.

Bradski, G. R. (1998). Computer Vision Face Tracking for Use in a Perceptual User Interface. Intel Technology Journal.